

Tianyu Chen

☎ (+1) 773-431-6690

✉ tianyuchen@utexas.edu

🔗 Google Scholar

🔗 TianyuCodings

EDUCATION

University of Texas at Austin	Ph.D. in Statistics (GPA: 4.0), Austin, TX	Oct. 2023 – July 2028 (expected)
University of Chicago	M.S. in Statistics (GPA: 3.97), Chicago, IL	Oct. 2021 – July 2023
Fudan University	B.S. in Statistics (GPA: 3.7), Shanghai, China	Sept. 2017 – July 2021

INDUSTRY EXPERIENCE

Microsoft Research | MAI Superintelligence

Research Scientist Intern (Mentors: Zhendong Wang, Lijuan Wang)

Feb. 2025 – Present

- Developed an object-centric vision-language model (VLM) agent that automates fine-grained, multi-turn image-editing evaluation, eliminating reliance on human annotators for benchmark construction. The project received media coverage in well-known Chinese outlets and is showcased on the project website.
- Led large language model (LLM) post-training research on mathematical reasoning, LLM-as-a-judge alignment, and on-policy-distillation, including recipe design for **autoregressive** and **diffusion-based** language models.

MORE INFORMATION

LinkedIn: <https://www.linkedin.com/in/tianyu-chen-1a056a160/>

Website: <https://tianyucodings.github.io/>

Research Interests: LLM post-training (RL and on-policy-distillation), image editing, and model distillation

SELECTED PUBLICATION HIGHLIGHTS

- **High-Impact Research Output:** Authored **12 papers** in the first **2.5** years of Ph.D. studies (**11 as first author**), with **9 published** in premier venues (NeurIPS, ICLR, AISTATS, PNAS), including a **NeurIPS Spotlight**.
- **Generative AI & LLM Alignment:** Pioneered frameworks for LLM post-training (reasoning & alignment) and automated evaluation (EdiVal-Agent), accepted at **ICLR 2026** and **NeurIPS 2025**.

GENERATIVE MODEL & REINFORCEMENT LEARNING

- **[ICML 2026 Submission] Tianyu Chen***, Yasi Zhang*, Mingyuan Zhou, Oscar Leong, Ying Nian Wu, Michal Lukasik[†]. “REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge.” (*Collaboration with Google DeepMind*).
- **[Neurips 2024] Tianyu Chen**, Zhendong Wang, Mingyuan Zhou. “Diffusion Policies creating a Trust Region for Offline Reinforcement Learning.” Published in: *Neurips 2024*. [\[arXiv\]](#). 2024.
- **[ICLR 2026] Tianyu Chen***, Yasi Zhang*, Zhi Zhang, Peiyu Yu, Shu Wang, Zhendong Wang, Kevin Lin, Xiaofei Wang, Zhengyuan Yang, Linjie Li, Chung-Ching Lin, Jianwen Xie, Oscar Leong, Lijuan Wang, Ying Nian Wu, Mingyuan Zhou. “EdiVal-Agent: An Object-Centric Framework for Automated, Scalable, Fine-Grained Evaluation of Multi-Turn Editing.” Accepted to: *ICLR 2026*. [\[OpenReview\]](#). 2026.
- **[ICLR 2026] Tianyu Chen***, Yasi Zhang*, Zhendong Wang, Ying Nian Wu, Mingyuan Zhou. “Score Distillation Beyond Acceleration: Generative Modeling from Corrupted Data.” Accepted to: *ICLR 2026*. [\[OpenReview\]](#). 2026.
- **[ICML 2026 Submission] Tianyu Chen***, Xinran Song*, Mingyuan Zhou. “A Generative Framework for Causal Estimation via Importance-Weighted Diffusion Distillation.” Preprint [\[arXiv\]](#). 2025.
- **[Neurips 2025] Yifan Sun**, Jingyan Shen, Yibin Wang, **Tianyu Chen**, Zhendong Wang, Mingyuan Zhou, Huan Zhang. “Improving Data Efficiency for LLM Reinforcement Fine-tuning Through Difficulty-targeted Online Data Selection and Rollout Replay.” Accepted to: *Neurips 2025*. [\[arXiv\]](#). 2025.
- **[PNAS] Tianyu Chen***, Jin-Hong Du*, Ming Gao, Jingshu Wang. “Model-based trajectory inference for single-cell rna sequencing using deep learning with a mixture prior.” Published in: *PNAS*. [\[PNAS\]](#). 2024.

STATISTICS

- **[Neurips 2025 Spotlight] Tianyu Chen**, Vansh Bansal, James G. Scott. “CoLT: The conditional localization test for assessing the accuracy of neural posterior estimates.” **Spotlight at: Neurips 2025**. [\[arXiv\]](#). 2025.
- **[ICML 2026 Submission] Tianyu Chen**, Vansh Bansal, James G. Scott. “The surprising strength of weak classifiers for validating neural posterior estimates.” Submitted to: *ICML 2026*. [\[arXiv\]](#). 2025.
- **[AISTATS 2025] Tianyu Chen**, James G. Scott. “Conditional diffusions for neural posterior estimation.” [\[arXiv\]](#). 2025.
- **[Neurips 2024] Tianyu Chen**, Kevin Bello, Francesco Locatello, Bryon Aragam, Pradeep Ravikumar. “Identifying General Mechanism Shifts in Linear Causal Representations.” Published in: *Neurips 2024*. [\[arXiv\]](#). 2024.
- **[Neurips 2023] Tianyu Chen**, Kevin Bello, Bryon Aragam, Pradeep Ravikumar. “iSCAN: Identifying Causal Mechanism Shifts among Nonlinear Additive Noise Models.” Published in: *Neurips 2023*. [\[arXiv\]](#). 2023.